Quantification: 536-63x_Hyp_RD_z15c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	10
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2859.2	2611.5	1055.85	px^2
major_axis_length_px	lacuna shape	109.847	113.675	31.5056	px
minor_axis_length_px	lacuna shape	36.291	36.1572	4.7547	px
aspect_ratio	lacuna shape	3.0307	3.3805	0.8272	unitless
eccentricity	lacuna shape	0.9243	0.9551	0.0585	unitless
solidity	lacuna shape	0.8145	0.8086	0.0547	unitless
roots_count	headline	10	8.5	3.3993	count
ring_length_r30_px	headline	351.3	344.5	98.795	px
ring_length_r60_px	reported	862.6	869	252.545	px
owned_length_px	ownership-dependent	1690.81	1463.09	1029.83	px
edge_count	ownership-dependent	77.9	63	50.3465	count
mean_edge_length_px	ownership-dependent	22.1305	21.8179	2.7296	px
perimeter_px	normalised	320.57	340.105	90.3594	px
ring_area_r30_px2	normalised	10797.3	11051	2044.83	px^2
ring_area_r60_px2	normalised	25748.4	25946	3580.52	px^2
in_frame_fraction_r30	normalised	0.9949	1	0.0162	unitless
in_frame_fraction_r60	normalised	0.9787	1	0.0673	unitless
ring_density_r30	normalised	0.0324707	0.0322383	0.0062323	px^-1
ring_density_r60	normalised	0.0332362	0.0301879	0.00779581	px^-1
roots_per_100px_perimeter	normalised	3.1831	3.1231	0.7618	per 100 px
ring_attached_length_r30_px	network v2	315.9	296.5	111.054	px
ring_attached_length_r60_px	network v2	728.4	714	248.311	px
ring_length_w_r30_px	network v2	397.334	394.969	113.636	px
ring_length_w_r60_px	network v2	978.631	996.749	286.219	px
sholl_crossings_r10	network v2	8.9	9	2.6854	count
sholl_crossings_r20	network v2	11.6	11.5	3.6576	count
sholl_crossings_r30	network v2	12	11.5	3.0551	count
loop_count	loops, report only	5	2.5	5.0553	count
loop_edge_share	loops, report only	0.1933	0.164	0.1575	share
ring_width_mean_r30_px	width	7.5396	7.7337	0.6903	px

Per field
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measure	value	units
analysed_area_px2	963248	px^2
total_skeleton_length_px	25658	px
canalicular_length_density_per_px	0.026637	px^-1
junction_count	492	count
junction_density_per_px2	0.00051077	px^-2
lacuna_count	10	count

lacunae_per_px2	1.038e-05	px^-2
skeleton_component_count	339	count
mean_component_length_px	75.6873	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0302436	px^-1
field_density_without_flagged_per_px	0.0251518	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	573	px
unattached_component_count	28	count
canalicular_length_density_attached_per_px	0.026637	px^-1
field_length_density_with_unattached_per_px	0.0272318	px^-1
excluded_area_px2	56736	px^2
analysed_field_area_px2	991840	px^2
loop_count	60	count
loop_edge_share	0.1807	share
edge_count	1284	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7296	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25663	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2274	76.8735	39.9156	1.9259	0.8546	0.8956	-0.3511	155.297	808.682	False
2	4398	148.873	40.64	3.6632	0.962	0.794	-0.278	209.417	635.271	False
3	4419	146.599	40.48	3.6215	0.9611	0.8651	-0.1486	344.271	977.852	False
4	2155	110.451	31.8217	3.4709	0.9576	0.7322	-0.4611	379.614	263.714	False
5	3362	116.899	43.0956	2.7125	0.9296	0.7535	-0.3179	479.939	367.637	False
6	1029	48.7214	29.7491	1.6377	0.7919	0.8654	-0.7175	487.359	818.536	False
7	2767	117.614	33.0604	3.5575	0.9597	0.7731	-0.1563	645.45	527.102	False
8	3423	136.956	33.4698	4.0919	0.9697	0.856	-0.2992	677.154	397.595	False
9	2309	104.739	31.8331	3.2902	0.9527	0.7872	-0.2648	715.611	173.698	False
10	2456	90.7499	38.8445	2.3362	0.9038	0.8231	-0.397	792.997	804.486	False

Network per lacuna

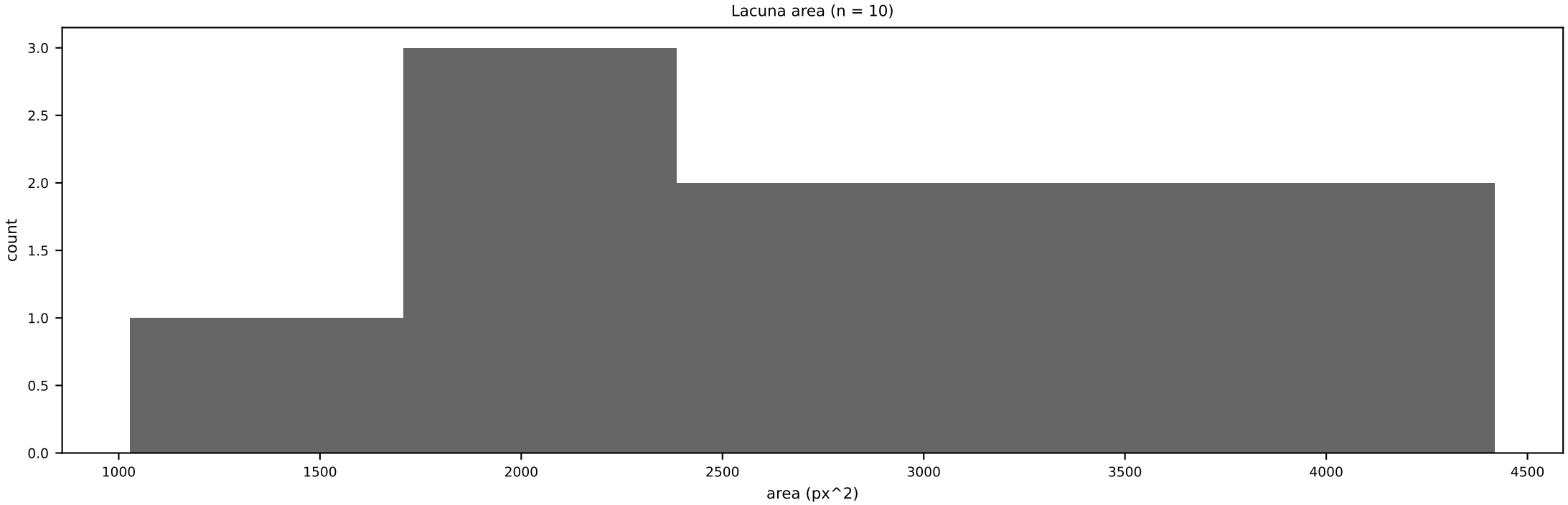
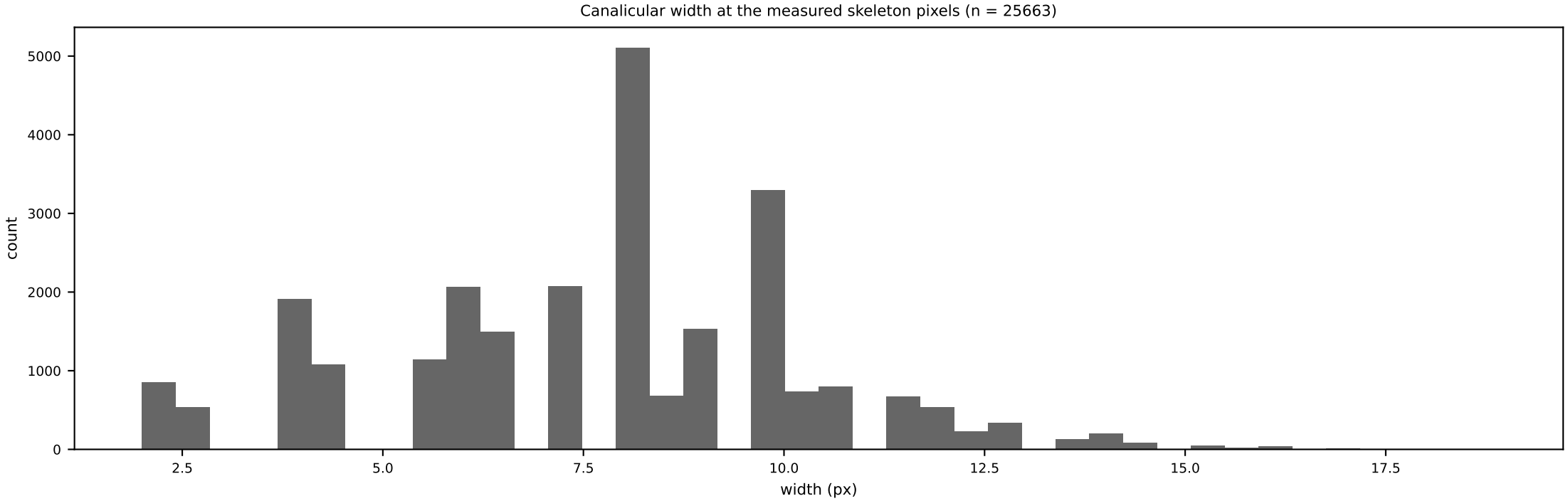
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	222	526	1453.57	58	25.0615	8.272
2	16	410	977	3962.43	189	20.9652	7.7857
3	9	323	759	858.637	41	20.9424	7.8921
4	8	398	1138	2403.1	106	22.6707	7.6816
5	13	534	1281	2518.32	128	19.6744	7.9372
6	6	225	532	642.671	33	19.4749	7.602
7	9	291	761	827.406	34	24.3355	6.8875

8	15	445	981	1472.61	81	18.1804	8.3703
9	8	366	995	1811.62	68	26.6414	6.6281
10	8	299	676	957.708	41	23.3587	6.3399

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	258.593	8854	23227	1	1	0.0250734	0.0226461	3.0937
2	480.115	14058	33319	1	1	0.0291649	0.0293226	3.3325
3	360.777	12077	24442	0.9489	0.7871	0.026745	0.0310531	2.4946
4	293.363	10490	25563	1	1	0.0379409	0.0445175	2.727
5	371.789	12101	27675	1	1	0.0441286	0.0462873	3.4966
6	142.61	6775	19178	1	1	0.0332103	0.0277401	4.2073
7	364.475	11610	26508	1	1	0.0250646	0.0287083	2.4693
8	326.25	11953	26656	1	1	0.0372292	0.0368022	4.5977
9	353.961	10492	26329	1	1	0.0348837	0.037791	2.2601
10	253.764	9563	24587	1	1	0.0312663	0.0274942	3.1525

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	141	446	238.61	586.375	4	7	8
2	387	899	461.99	1112.59	10	13	15
3	285	530	371.777	871.566	10	11	9
4	374	926	450.747	1274.49	12	13	13
5	509	1192	611.215	1462.83	13	18	18
6	206	503	256.622	606.701	6	7	11
7	249	614	335.534	893.377	7	8	10
8	436	876	494.848	1100.12	9	16	12
9	308	814	418.161	1124.28	9	11	14
10	264	484	333.836	753.985	9	12	10

536-63x_Hyp_RD_z15c1-2: distributions, pre-validation, px